Cambridge Assessment International Education

Cambridge O Level

ADDITIONAL MATHEMATICS **4037/21**

Paper 2 **May/June 2022**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt answers which round to
cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
nfww not from wrong working
oe or equivalent
rot rounded or truncated
SC Special Case
soi seen or implied

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Question	Answer	Marks	Partial Marks
1(a)	$w-1=\log_5 12$ or $w-1=\dfrac{\log 12}{\log 5}$	**M1**	
	$w=2.54$ cao	**A1**	
1(b)	Rewrites in quadratic form e.g.: $y=x^{\frac{1}{3}}\quad y^2-5y+6=0$ or $\left(x^{\frac{1}{3}}\right)^2-5x^{\frac{1}{3}}+6=0$	**M1**	
	and factorises or solves e.g. : $(y-2)(y-3)=0$	**M1**	Factorising *their* 3 term quadratic
	$x=8$ and $x=27$	**A1**	
2(a)	$\lg\dfrac{x^2}{3(x+6)}$ oe, nfww	**B2**	**B1** for any two log laws applied correctly e.g. $\lg\dfrac{x^2}{x+6}-\lg 3$
2(b)	$\lg\dfrac{x^2}{3(x+6)}=\lg 1$ or $10^0=\dfrac{x^2}{3(x+6)}$	**B1**	FT *their* $\lg\dfrac{x^2}{3(x+6)}$ providing a single logarithm
	$x^2-3x-18=0$	**B1**	**dep** on **B2** in **part (a)**
	Factorises or solves their 3-term quadratic	**M1**	
	$x=6$ indicated as only solution	**A1**	**dep** on **B2** in **part (a)**
3	Valid method to find m $m=\dfrac{8-1}{9-16}\,[=-1]$	**M1**	
	Valid method to find c e.g. $1=their(-1)\times 16+c$	**M1**	**FT** *their m*
	$\sqrt[3]{y}=their(-1)x^2+their17$	**A1**	Equation with correct variables and $\sqrt[3]{y}=$
	$y=\left(-x^2+17\right)^3$ oe, isw	**A1**	

Question	Answer	Marks	Partial Marks
4	$27m - 153 + 3n + 6 = 0$ or better	**B1**	
	$-m - 17 - n + 6 = -12$ or better	**B1**	
	Eliminates one unknown for a pair of linear equations in m and n and solves for one unknown	**M1**	
	$m = 6$, $n = -5$	**A2**	**A1** for either
	-24 cao	**A1**	
5(a)(i)	$1 + 4nx + 8n(n-1)x^2$	**B2**	**B1** for any two correct terms or all 3 correct but listed not summed
5(a)(ii)	$8n(n-1) - 16n$	**M1**	**FT** from (i) identifying correct terms and combining *their* coefficient of $x^2 - 4 \times$ *their* coefficient of x
	Solves or factorises *their* 3-term quadratic in n only	**M1**	Forms a 3-term quadratic = 0 and solves except allow '= a constant' if they go on to complete to square
	$n = 29$ only	**A1**	
5(b)	${}^{10}C_2 \times \left(\frac{x}{2}\right)^8 \times \left(-\frac{8}{x^4}\right)^2$ soi	**M1**	Must be clearly identified not in expansion
	11.25 or $\frac{45}{4}$	**A1**	
6(a)(i)	$6 \times 6 \times 5 \times 4 \times 3$ oe	**M1**	
	2160	**A1**	
6(a)(ii)	Full correct calculation (360 + 900) 'how many end with 0'+ 'how many end with 2, 4, 6' oe	**M2**	**M1** for any correct product soi Eg $(6 \times 5 \times 4 \times 3 \times 1)$ or 360 $(5 \times 5 \times 4 \times 3 \times 1)$ or 300 $(5 \times 5 \times 4 \times 3 \times 3)$ or 900
	1260 cao	**A1**	
6(b)	${}^{15}C_7 - {}^{9}C_7$	**M1**	
	6399	**A1**	

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Question	Answer	Marks	Partial Marks
6(c)(i)	$\frac{n!}{(n-3)!3!}+\frac{n!}{(n-2)!2!}$	**B2**	**B1** for either expression correct
	$\frac{n(n-1)(n-2)}{6}+\frac{n(n-1)}{2}$	**M2**	**M1** for either expression correct
	$\frac{n(n-1)(n-2+3)}{6}$ or $\frac{n^3-3n^2+2n}{6}+\frac{3n^2-3n}{6}$ leading to $\frac{1}{6}(n^3-n)$	**A1**	
6(c)(ii)	$n(n^2-25)=0$ oe	**M1**	
	$n = 5$ as the only solution	**A1**	
7	$\frac{\mathrm{d}}{\mathrm{d}x}(\sin 3x)=3\cos 3x$ soi	**B1**	
	$u=(1+\sin 3x)^4$ $\frac{\mathrm{d}u}{\mathrm{d}x}=4(1+\sin 3x)^3(3\cos 3x)$ soi	**M1**	**FT** *their* $3\cos 3x$
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\sqrt{x}\left(4(1+\sin 3x)^3(3\cos 3x)\right)-(1+\sin 3x)^4\times\frac{1}{2}x^{-\frac{1}{2}}}{\left(\sqrt{x}\right)^2}$	**M1**	FT *their* $\frac{\mathrm{d}u}{\mathrm{d}x}$ *or their* $\frac{\mathrm{d}v}{\mathrm{d}x}$ but not both
	Correct derivative	**A1**	
	Evaluates *their* derivative at $x = 1.9$ and multiplies by h	**M1**	
	$0.651h$	**A1**	
8(a)	$x=6\sqrt{3}\sin 60 \quad y=-6\sqrt{3}\cos 60$ oe and completion to $9\mathbf{i}-3\sqrt{3}\mathbf{j}$	**B2**	**B1** for either x or y correct Allow **SC1** for verification that $9\mathbf{i}-3\sqrt{3}\mathbf{j}$ has a bearing of 120° and that $9\mathbf{i}-3\sqrt{3}\mathbf{j}$ has a magnitude of $6\sqrt{3}$
8(b)	$(5\mathbf{i}+16\mathbf{j})+3\left(9\mathbf{i}-3\sqrt{3}\mathbf{j}\right)$ oe, isw	**B1**	
8(c)	$29\mathbf{i}+16\mathbf{j}+t(-12\sqrt{3}\mathbf{j})$ oe, isw	**B1**	

Question	Answer	Marks	Partial Marks
8(d)	Forms $\overrightarrow{AB}$ or $\overrightarrow{BA}$ when $t = 1$ e.g. $\overrightarrow{BA} = (32\mathbf{i} + (16 - 9\sqrt{3})\mathbf{j}) - (29\mathbf{i} + (16 - 12\sqrt{3})\mathbf{j})$ oe	**B1**	**FT** *their* (b) and (c) with $t = 1$
	$\sqrt{3^2 + (3\sqrt{3})^2}$	**M1**	**FT** *their* $\overrightarrow{AB}$ or $\overrightarrow{BA}$
	6 (km)	**A1**	cao
9(a)	$h = \sqrt{x^2 - \left(\frac{x}{2}\right)^2}$ oe or $\cos 30 = \frac{h}{x}$ oe or $\frac{1}{2}xh = \frac{1}{2}x^2 \sin 60$	**M1**	Correct expression connecting x and h
	$x = \frac{2h}{\sqrt{3}}$ oe	**A1**	Must be $x =$
	$V = \frac{1}{2} \times \left(\textit{their}\ \frac{2h}{\sqrt{3}}\right)^2 \times \sin 60° \times 5$ or $\frac{1}{2} \times h \times \textit{their}\ \frac{2h}{\sqrt{3}} \times 5$	**M1**	
	Correct completion to given answer $V = \frac{5\sqrt{3}\,h^2}{3}$	**A1**	
9(b)	Correct derivative of V e.g. $\frac{10\sqrt{3}}{3}h$	**B1**	
	$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}t} \times \frac{\mathrm{d}h}{\mathrm{d}V}$ soi	**B1**	Not $\frac{\partial V}{\partial h}$
	$0.5 \div \left(\frac{10\sqrt{3}}{3} \times 0.1\right)$ oe	**M1**	
	0.866 (metres per minute) or 0.8660[25…] rot to 4 or more sf	**A1**	Allow $\frac{\sqrt{3}}{2}$ isw
10(a)	$\ln x - 1$	**B2**	**B1** for $[1\times]\ln x + x \times \frac{1}{x}[-2]$ oe

Question	Answer	Marks	Partial Marks
10(b)	$x+\frac{1}{x}+2$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{x^2}{2}+\ln x+2x(+c)$	**M2**	**M1** for $\frac{\mathrm{d}y}{\mathrm{d}x}=...+\ln x+...$ or for $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{x^2}{2}+...+2x$
	Substitution to find c: $\frac{\mathrm{e}^2}{2}+2\mathrm{e}=\frac{\mathrm{e}^2}{2}+\ln\mathrm{e}+2\mathrm{e}+c$ $[c=-1]$	**M1**	**FT** *their* attempt to integrate dependent on at least **M1**
	$y=\int\left(\frac{x^2}{2}+2x+\ln x-1\right)\mathrm{d}x$	**A1**	
	Integrates and uses (a) $y=\frac{x^3}{6}+\frac{2x^2}{2}+x\ln x-2x+C$	**M1**	Dependent on **M2M1** **FT** error in c only
	$\frac{\mathrm{e}^3}{6}+\mathrm{e}^2=\frac{\mathrm{e}^3}{6}+\mathrm{e}^2+\mathrm{e}\ln\mathrm{e}-2\mathrm{e}+C$	**M1**	Substitution to find C Dependent on previous **M1**
	$y=\frac{x^3}{6}+x^2+x\ln x-2x+\mathrm{e}$	**A1**	
11	$B\left(\frac{\pi}{10},0\right)$ soi	**B1**	
	$\int_0^{\frac{\pi}{4}}\mathrm{e}^{\frac{x}{2}}\,\mathrm{d}x-\int_0^{\frac{\pi}{10}}\cos 5x\,\mathrm{d}x$	**M2**	**M1** Integrates $\mathrm{e}^{\frac{x}{2}}$ to $k\mathrm{e}^{\frac{x}{2}}$, $k\neq 0$ **M1** Integrates $\cos 5x$ to $k\sin 5x$, $k>0$ or $k=-\frac{1}{5}$
	$\left[2\mathrm{e}^{\frac{x}{2}}\right]_0^{\frac{\pi}{4}}-\left[\frac{\sin 5x}{5}\right]_0^{\frac{\pi}{10}}$	**A2**	**A1** for each part correct
	$\left[2\mathrm{e}^{\frac{\pi}{8}}-2\mathrm{e}^0\right]-\left[\frac{1}{5}\sin\frac{5\pi}{10}-\frac{1}{5}\sin 0\right]$	**M1**	**M1** for correct attempt at subtraction and for one correct use of limits dependent on at least **M1**for integration and **B1**
	$2\mathrm{e}^{\frac{\pi}{8}}-\frac{11}{5}$ isw	**A1**	

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